CardioVision AI

Multi-Vessel Risk Assessment & Clinical Vector Evaluation

RESEARCH PROTOTYPE EVALUATION REPORT — NOT FOR CLINICAL DIAGNOSIS

Evaluation Date: Tue, Oct 13, 2026, 09:17 PM
Patient ID: CV-2026-8492
Architecture: 57-Feature Four-Model Architecture
Active Mode: Experimental Candidate Mode

PATIENT NAME	PATIENT ID	AGE / BIOLOGICAL SEX	HEIGHT / WEIGHT / BMI
Alex Morgan (Demo Patient)	**CV-2026-8492**	**58 yrs · Male**	**170 cm / 72 kg (24.9 kg/m²)**

1. Cardiovascular Prediction Summary

Multi-vessel predictive classification derived from the 57-dimensional clinical feature vector.

OVERALL CAD	LAD ARTERY	LCX ARTERY	RCA ARTERY
91.7%	**79.0%**	**58.0%**	**40.4%**
CAD	**Stenotic**	**Stenotic**	**Stenotic**
cad_model.pkl Threshold: **50.0%**	lad_model.pkl Threshold: **50.0%**	lcx_model_candidate.pkl Threshold: **45.0%**	rca_model_candidate.pkl Threshold: **40.0%**
High Est. Prob (≥70%)	**High Est. Prob (≥70%)**	**Moderate Est. Prob (50–69%)**	**Low Est. Prob (<50%)**

Model Estimation Notice: Values are model-estimated statistical classification probabilities, not clinically calibrated epidemiological risk scores or measured anatomical luminal stenosis narrowing percentages.

2. Clinical Input Measurements Summary

Patient clinical parameters from active evaluation session evaluated against informational reference benchmarks.

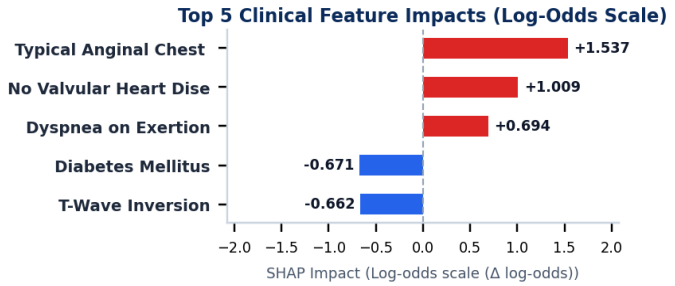
Clinical Parameter	Patient Value	General Reference Benchmark	Informational Evaluation Status
Resting Systolic BP (BP)	**130 mmHg**	< 120 mmHg (resting systolic)	Above reference (≥ 130 mmHg, Informational)
Resting Pulse Rate (PR)	**70 bpm**	60 – 100 bpm	Within standard resting range
Left Ventricular EF (EF-TTE)	**50%**	≥ 50%	LVEF ≥ 50% (Standard systolic function)
Fasting Blood Sugar (FBS)	**98 mg/dL**	70 – 99 mg/dL	Within fasting range (70–99 mg/dL)
Serum Creatinine (CR)	**1.0 mg/dL**	0.7 – 1.3 mg/dL	Within displayed adult reference range (Informational)
LDL Cholesterol	**100 mg/dL**	< 100 mg/dL	Above standard reference (≥ 100 mg/dL, Informational)
HDL Cholesterol	**39 mg/dL**	> 40 mg/dL (M) / > 50 mg/dL (F)	Below reference (< 40 mg/dL, Informational)
Triglycerides (TG)	**122 mg/dL**	< 150 mg/dL	Within reference (< 150 mg/dL)
Evaluated Clinical Vector	**57 Model Inputs**	Standardized 7-domain vector	34 numeric + 23 binary clinical indicators synchronized

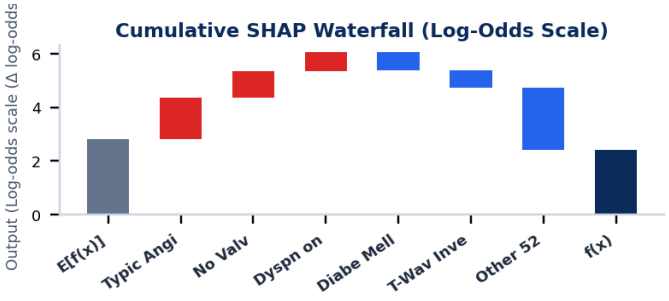
3. Patient-Specific SHAP Explainability (CAD & LAD)

Feature contributions calculated via game-theoretic Shapley values. **Scale Note:** TreeExplainer outputs additive probability contributions (Δ probability). LinearExplainer outputs additive log-odds contributions (Δ log-odds). Scales are strictly separated.

CAD Model Explainability (cad_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **CAD** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **2.810 log-odds** | Final Patient Output f(x): **2.407 log-odds** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

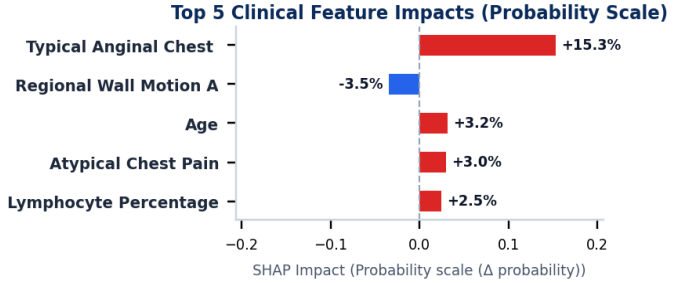
Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	**Typical Anginal Chest Pain**	Symptoms	**Present (1)**	**+1.537 log-odds**	**Increases Risk**
#2	**No Valvular Heart Disease**	Valvular Disease	**Present (1)**	**+1.009 log-odds**	**Increases Risk**
#3	**Dyspnea on Exertion**	Symptoms	**Absent (0)**	**+0.694 log-odds**	**Increases Risk**
#4	**Diabetes Mellitus**	Medical History	**Absent (0)**	**-0.671 log-odds**	**Decreases Risk**
#5	**T-Wave Inversion**	ECG & Telemetry	**Absent (0)**	**-0.662 log-odds**	**Decreases Risk**

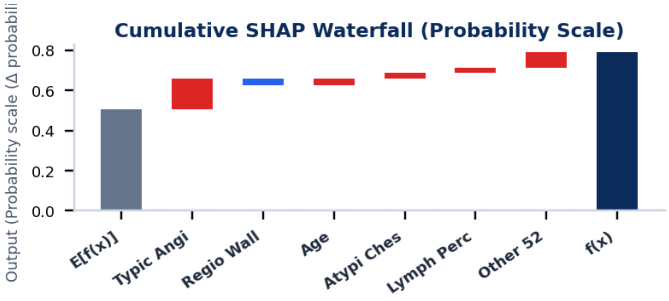




LAD Artery Model Explainability (lad_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.7%** | Final Patient Output f(x): **79.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	**Typical Anginal Chest Pain**	Symptoms	**Present (1)**	**+15.3%**	**Increases Risk**
#2	**Regional Wall Motion Abnormality**	Echocardiography	**Absent (0)**	**-3.5%**	**Decreases Risk**
#3	**Age**	Basic Demographics	**58 years**	**+3.2%**	**Increases Risk**
#4	**Atypical Chest Pain**	Symptoms	**Absent (0)**	**+3.0%**	**Increases Risk**
#5	**Lymphocyte Percentage**	Biomarkers	**32 %**	**+2.5%**	**Increases Risk**



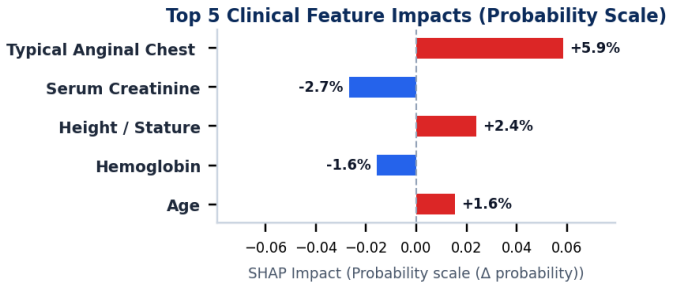


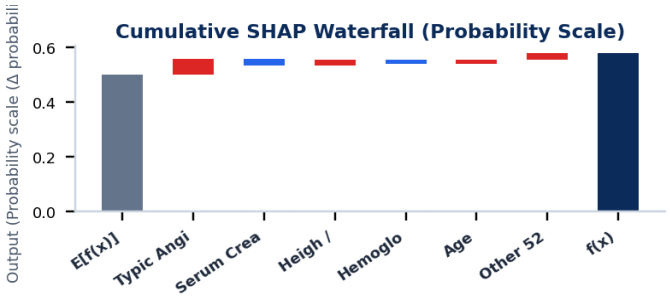
3. Patient-Specific SHAP Explainability (LCX & RCA)

Candidate and baseline branch vessel models with explicit scale separation and additivity verification.

LCX Artery Model Explainability (lcx_model_candidate.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic (Class 1)** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.1%** | Final Patient Output f(x): **58.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

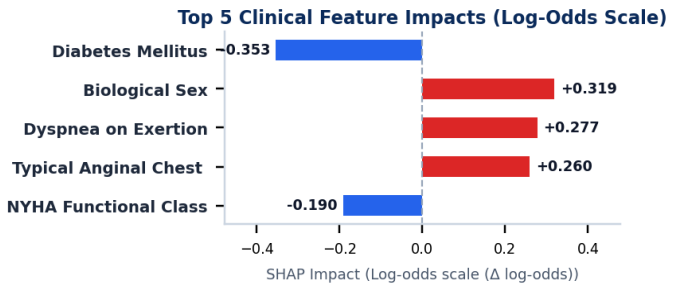
Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	Typical Anginal Chest Pain	Symptoms	Present (1)	+5.9%	Increases Risk
#2	Serum Creatinine	Biomarkers	1 mg/dL	-2.7%	Decreases Risk
#3	Height / Stature	Basic Demographics	170 cm	+2.4%	Increases Risk
#4	Hemoglobin	Biomarkers	13.2 g/dL	-1.6%	Decreases Risk
#5	Age	Basic Demographics	58 years	+1.6%	Increases Risk

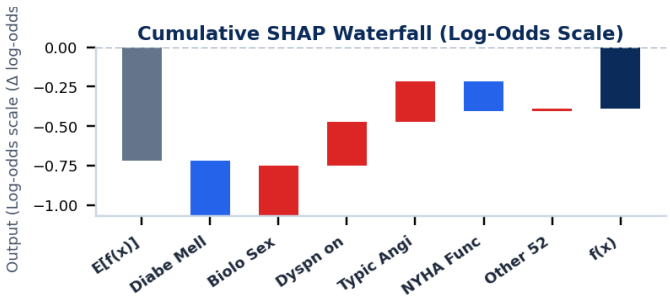




RCA Artery Model Explainability (rca_model_candidate.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **Stenotic (Class 1)** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **-0.717 log-odds** | Final Patient Output f(x): **-0.389 log-odds** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	Diabetes Mellitus	Medical History	Absent (0)	-0.353 log-odds	Decreases Risk
#2	Biological Sex	Basic Demographics	Male	+0.319 log-odds	Increases Risk
#3	Dyspnea on Exertion	Symptoms	Absent (0)	+0.277 log-odds	Increases Risk
#4	Typical Anginal Chest Pain	Symptoms	Present (1)	+0.260 log-odds	Increases Risk
#5	NYHA Functional Class	Physical Findings	Class 0	-0.190 log-odds	Decreases Risk





4. 3D Coronary Heart Visualization

High-resolution anatomical perspective from the interactive Three.js 106-mesh BodyParts3D cardiac model.



5. Model Performance and Reliability

Experimental Mode: Exploratory nested threshold cross-validation metrics for candidate models. Never claim independent prospective clinical validation.

Model	Target Outcome	Eval Framework	Macro F1 / F1	Stenotic Recall	Precision	Bal. Accuracy	Accuracy	ROC-AUC
CAD Model (cad_model.pkl)	Overall CAD	Held-out test cohort	89.7% (F1)	N/A	N/A	N/A	**85.2%**	**0.917**
LAD Model (lad_model.pkl)	LAD Stenosis	Held-out test cohort	67.7% (F1)	N/A	N/A	N/A	**65.6%**	**0.756**
LCX Candidate (lcx_model_candidate.pkl)	LCX Stenosis	**Exploratory nested CV**	**0.689 (Macro F1)**	**0.597**	**0.634**	**0.687**	*Not evaluated*	*Not evaluated*
RCA Candidate (rca_model_candidate.pkl)	RCA Stenosis	**Exploratory nested CV**	**0.651 (Macro F1)**	**0.632**	**0.545**	**0.657**	*Not evaluated*	*Not evaluated*

Candidate Cross-Validation Notice: LCX and RCA candidate models report exploratory nested threshold cross-validation metrics (LCX Macro F1: 0.689, Stenotic Recall: 0.597, Precision: 0.634, Balanced Accuracy: 0.687; RCA Macro F1: 0.651, Stenotic Recall: 0.632, Precision: 0.545, Balanced Accuracy: 0.657). Accuracy and ROC-AUC were not evaluated. These exploratory metrics must not be confused with or reported as Standard mode metrics.

6. Research Limitations, Educational Guidance & Disclaimer

Scope of Research Evaluation & Clinical Considerations:
- Model estimates represent statistical pattern matching against retrospective cohorts, not definitive clinical diagnoses.
- Results must always be interpreted in full clinical context by a licensed medical practitioner.
- Machine learning probabilities cannot substitute for direct coronary angiography, CT angiography, or stress imaging.
- Diagnostic and management decisions depend on patient presentation, physical exam, and comprehensive clinical history.

RESEARCH PROTOTYPE — NOT A CERTIFIED MEDICAL DEVICE:
This software and report are intended strictly for educational, scientific research, and methodology evaluation purposes. They are not approved or cleared by any regulatory health agency (e.g., FDA, CE, PMDA) for clinical diagnostic or therapeutic decision-making. Predictions and SHAP explainability analyses do not substitute for formal cardiovascular imaging, laboratory testing, or physician consultation.